



Universidade do Minho



Honey processing for protein-based authenticity rapid testing

Dimitrios Papakyriakou, Eleni Lamprou and Despina P. Kalogianni¹

Group of Analytical Chemistry & Nanotechnology, Department of Chemistry, University of Patras, GR26504, Rio, Patras, Greece

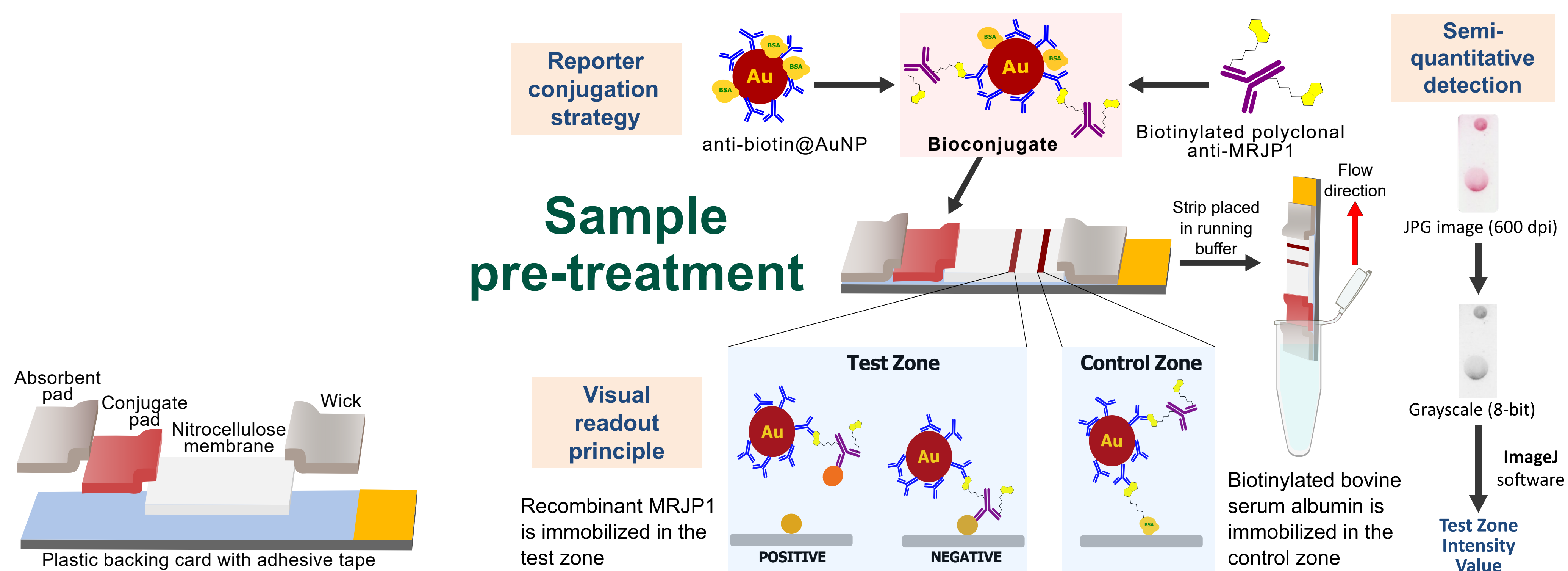
¹ kalogian@upatras.gr

Introduction

- Honey is a natural sweet substance produced by bees and processed into a food product with antibacterial, antioxidant, antimutagenic, and anti-inflammatory properties
- Adulteration with sugar syrups:
 - Diminishes nutritional value and food safety
 - Results in financial losses for the apiculture sector
- Major royal jelly protein 1 (MRJP1)**
 - Bee-derived protein
 - MRJP1 content in honey decreases upon adulteration
 - Potential honey authenticity marker

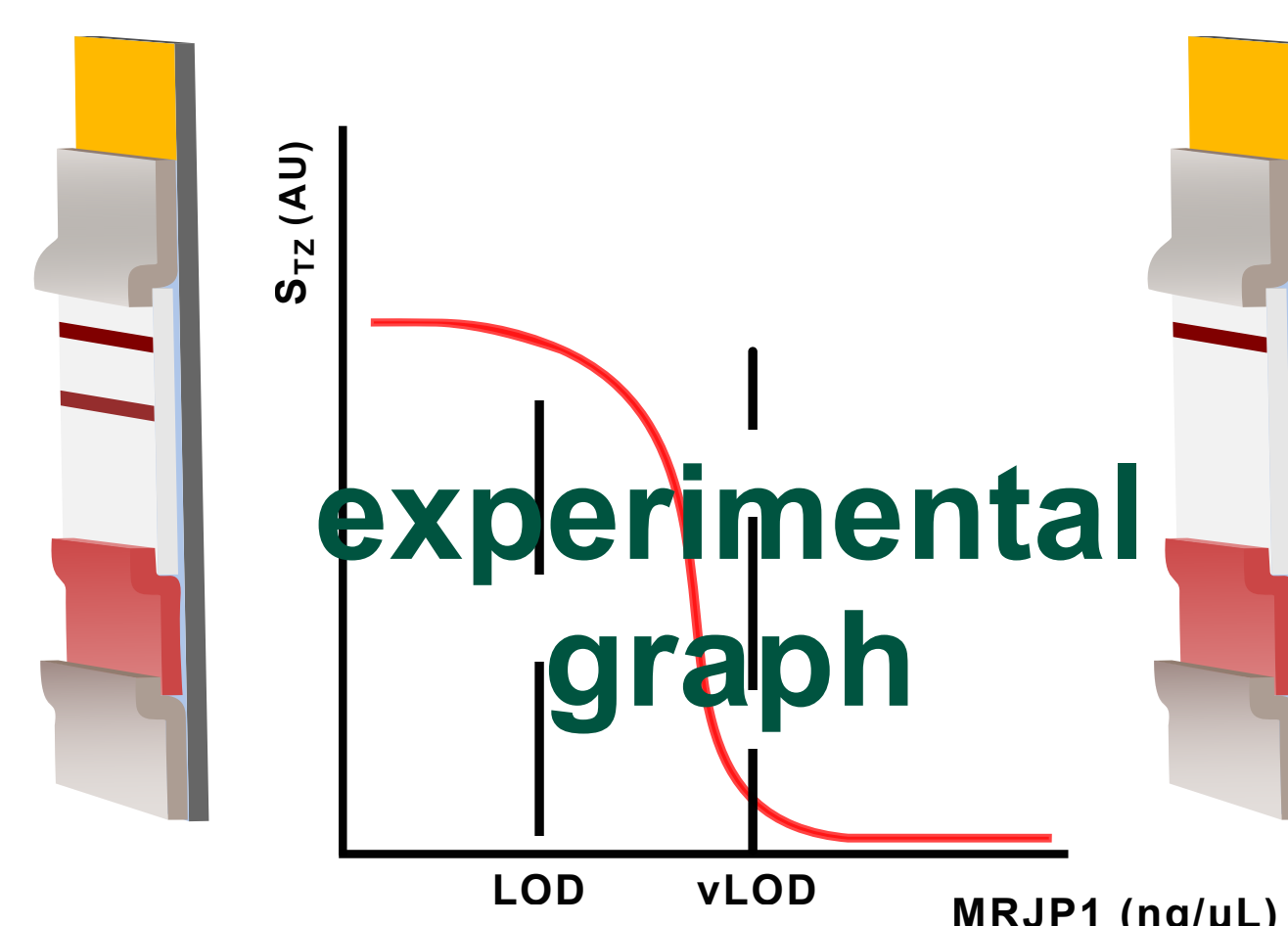
AIM:The development of a rapid assay for honey authenticity assessment.

- **Competitive lateral flow immunoassay for the detection of major royal jelly protein 1.**



Results

strip jpg sigmoid



Linear range calibration curve

Analytical performance metrics

Comparison graph

Performance among analytical techniques

Conclusions

simplicity, in-field application by non-expert users, with no instrumentation required

PDF available

